

An Overview of Space-Time and Space-Frequency Block Codes for Fading Multipath Channels

EE5141: Introduction to Wireless and Cellular Communication

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The Problem: Channel Selectivity and Deep Fades

- Multipath propagation and relative transceiver velocity give rise to a Fading Multipath Channel.
- **Time Selectivity:** Channel varies over time due to relative motion.
- **Frequency Selectivity:** Channel varies across the subcarrier frequencies.
- We cannot eliminate fading, but we can mitigate it using **Diversity**.

What is Diversity?

Definition: Diversity is the principle of transmitting multiple independent copies of the same signal over channels that experience independent fading.

Examples of Diversity:

- **Spatial Diversity:** Using multiple geographically separated antennas.
- **Time Diversity:** Transmitting replicas of the signal at different time intervals (e.g., interleaving).
- **Frequency Diversity:** Transmitting replicas on different subcarriers (e.g., OFDM).

Space-Time Block Codes (STBC) and **Space-Frequency Block Codes (SFBC)** provide spatial diversity on top of existing time/frequency diversity.

The Generalized Codeword Matrix (C): A block of N_s serial symbols is encoded into a matrix sent over M_t transmit antennas across T OFDM blocks (each with N subcarriers):

$$\mathbf{C}_{NT \times M_t} = \begin{bmatrix} \mathbf{c}_1^1 & \cdots & \mathbf{c}_{M_t}^1 \\ \vdots & \ddots & \vdots \\ \mathbf{c}_1^T & \cdots & \mathbf{c}_{M_t}^T \end{bmatrix}$$

- **STBC:** The coding dimension spans consecutive *time slots* (T).
- **SFBC:** The coding dimension spans adjacent *subcarriers* (N).

Coding Rate (R): $R = \frac{N_s}{NT}$ (Data Symbols / Channel Uses) - higher reduces the throughput.

The Alamouti Code

- An Orthogonal STBC - block code for 2 transmit antennas at full rate ($R = 1$).
- The transmission matrix is designed such that the inner product of the sequence transmitted from antenna 1 and antenna 2 is strictly zero.
- This orthogonality allows the code to be applied as an STBC (over time) or an SFBC (over OFDM subcarriers).

Mathematical Representation of Alamouti Codes

For $N_s = 2$ symbols (x_1, x_2) over $M_t = 2$ antennas and $NT = 2$ resources, achieving exactly $R = 1$:

$$\mathbf{X} = \begin{bmatrix} x_1 & x_2 \\ -x_2^* & x_1^* \end{bmatrix} \begin{array}{l} \leftarrow \text{Resource 1 } (t_1 \text{ or } f_1) \\ \leftarrow \text{Resource 2 } (t_2 \text{ or } f_2) \end{array}$$

Column 1 $\rightarrow$ Antenna 1, Column 2 $\rightarrow$ Antenna 2.

Orthogonality:

$$\mathbf{x}_1^H \mathbf{x}_2 = \begin{bmatrix} x_1^* & -x_2 \end{bmatrix} \begin{bmatrix} x_2 \\ x_1^* \end{bmatrix} = 0$$

Received Signals: Assuming a constant channel (h_1, h_2) across either time or frequency resources for Tx 1 and Tx 2 respectively:

$$y_1 = h_1 x_1 + h_2 x_2 + n_1$$

$$y_2 = -h_1 x_2^* + h_2 x_1^* + n_2$$

Effective Channel Matrix Representation ($\mathbf{H}_{\text{eff}}$):

$$\begin{bmatrix} y_1 \\ y_2^* \end{bmatrix} = \underbrace{\begin{bmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{bmatrix}}_{\mathbf{H}_{\text{eff}}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2^* \end{bmatrix}$$

A Nice Receiver!

Because $\mathbf{H}_{\text{eff}}$ has exactly orthogonal columns, multiplying by its Hermitian transpose yields a purely diagonal scalar matrix:

$$\mathbf{H}_{\text{eff}}^H \mathbf{H}_{\text{eff}} = \begin{bmatrix} h_1^* & h_2 \\ h_2^* & -h_1 \end{bmatrix} \begin{bmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{bmatrix} = \begin{bmatrix} |h_1|^2 + |h_2|^2 & 0 \\ 0 & |h_1|^2 + |h_2|^2 \end{bmatrix}$$

We use Maximal Ratio Combining (MRC) to combine the received signals:

$$\tilde{x}_1 = h_1^* y_1 + h_2 y_2^* = (|h_1|^2 + |h_2|^2) x_1 + n'_1$$

$$\tilde{x}_2 = h_2^* y_1 - h_1 y_2^* = (|h_1|^2 + |h_2|^2) x_2 + n'_2$$

This gives 2 independent, single-symbol scalar minimum distance problems (low-complexity ML receiver) with the output of the MRC:

$$\hat{x}_i = \arg \min_{x \in \mathcal{S}} |\tilde{x}_i - (|h_1|^2 + |h_2|^2) x|^2 \quad \text{for } i \in \{1, 2\}$$

Alamouti Transmission Scheme

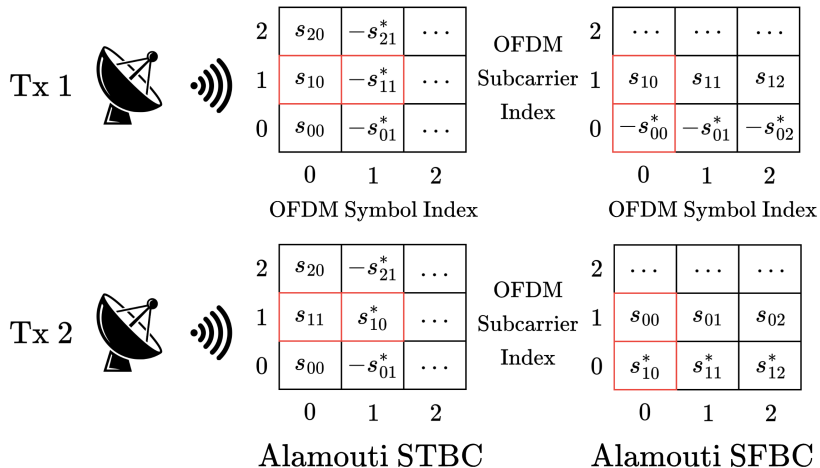


Figure 1: Orthogonal Mapping of Symbols in the Alamouti Scheme

Experimental Setup

We consider a transmitter with 2 sufficiently spaced antennas experiencing independent fading channels, compared against a baseline SISO OFDM system. To ensure a fair comparison, total transmit power is conserved (each Alamouti antenna transmits at $P/2$). CSI is used for the ML decoding formula given earlier.

Parameter	Value
Modulation	QPSK
Carrier Frequency (f_c)	3.5 GHz
Subcarrier Spacing	15 kHz
FFT Size / OFDM Symbols	64 / 64
Low / High Delay Spread	200 ns / 4.0 μ s
Low / High Doppler Spread	35 Hz (10.8 kmph) / 583.33 Hz (180 kmph)
E_b/N_0 Range	-1.0 to 20.0 dB

Simulation Environment

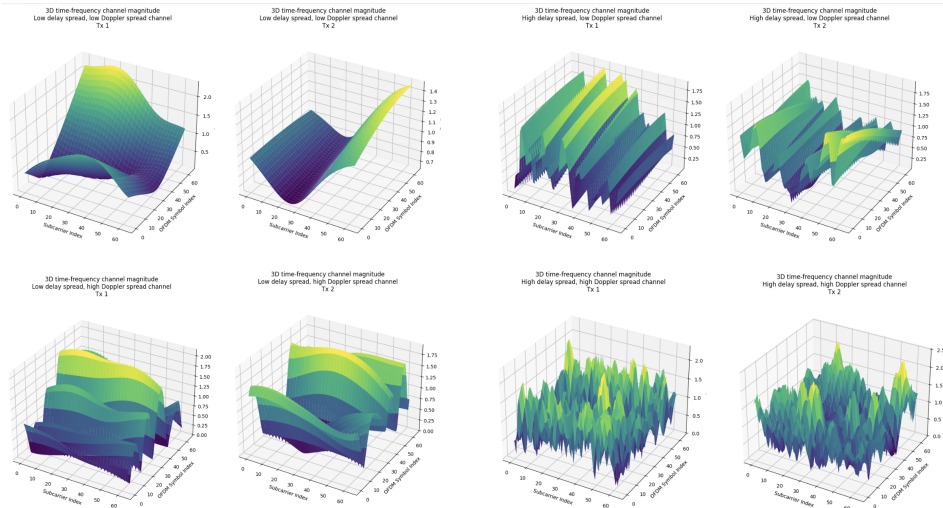


Figure 2: Diverse Channel environments to test STBC/SFBC

Results: Baseline BER Performance

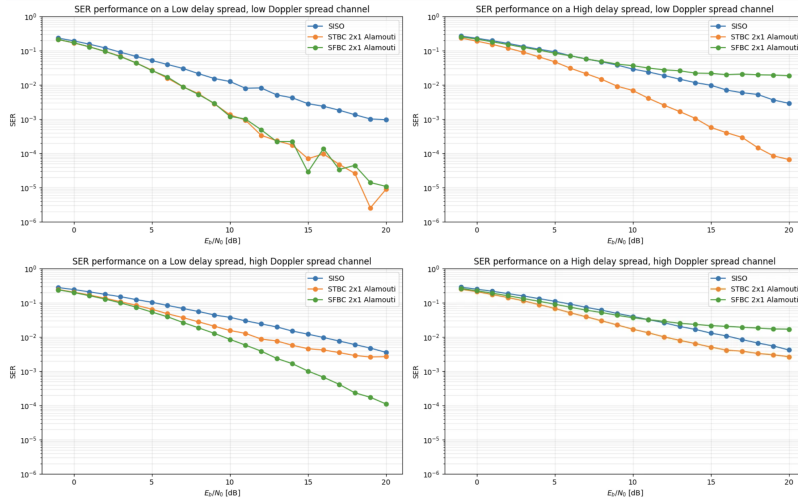


Figure 3: Symbol Error Rate (SER) vs E_b/N_0 demonstrating spatial diversity gain over the SISO baseline.

- Both STBC and SFBC maintain $R = 1$, so the data rate can be maintained.
But diversity gain improves the throughput.
- The equalized symbol is scaled by $(|h_1|^2 + |h_2|^2)$, which is the effective channel for a symbol - deep fade probability falls.

Diversity Gain across Fading Profiles:

1. **Low Delay, Low Doppler:** Both STBC and SFBC provide optimal diversity.
2. **High Doppler (Time-Selective):** STBC fails as $h(t_1) \neq h(t_2)$, breaking orthogonality. SFBC excels.
3. **High Delay Spread (Freq-Selective):** SFBC fails as $h(f_1) \neq h(f_2)$. STBC excels.
4. **High Doppler & High Delay:** Both orthogonal block codes suffer severe self-interference.

Adding Forward Error Correction (LDPC)

How do we correct for the higher channel selectivity with a low-complexity receiver?

Diversity Gain vs. Coding Gain:

- **Diversity Gain:** Alters the *slope* of the BER curve - large SNR analysis.
- **Coding Gain:** Alters the horizontal *shift* of the BER curve.

Implementation: We apply a Rate 0.5 LDPC code with a codeword length of 8192 bits (1 frame size). The bitstream is encoded, mapped, and then passed through the Alamouti OFDM modulator. Upon reception, we decode the frame using belief propagation on the LLRs generated after MRC equalization.

Results: Alamouti with LDPC Coding

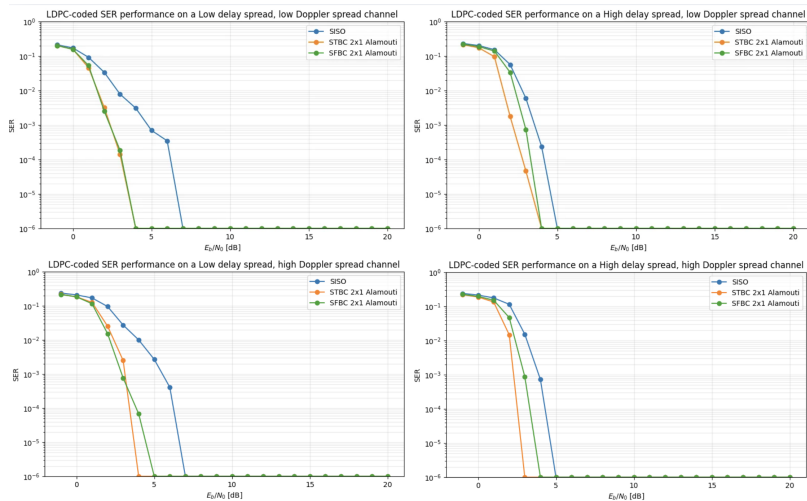


Figure 4: Combined robustness of coding gain (shift) and diversity gain (slope).

- The LDPC code provides some support to the weakness of STBC at high Doppler and SFBC at high Delay. Also, LDPC itself works well with frequency diversity.
- Coding and Diversity both complement each other - in MIMO OFDM.

- **Space-Time-Frequency Block Codes (STFBC):** For highly selective channels (high Doppler *and* high delay spread) - can visualize a 3D space of Space + Frequency + Time.
- **More than 2 Transmit Antennas:** Achieving perfect orthogonality for ≥ 3 antennas - Radon-Hurwitz matrix designs. But the cost is reduction in code rate.
- **Spatial Mapping Expansion:** 3D repeated stacking of a code matrix to achieve more diversity at the cost of bandwidth efficiency.
- **CSI Feedback** Switching between STBC and SFBC based on CSI feedback.

Key References

- **Reference 1:** S. M. Alamouti, "A simple transmit diversity technique for wireless communications," IEEE JSAC, 1998.
- **Reference 2:** K. F. Lee and D. B. Williams, "A space-frequency transmitter diversity technique for OFDM systems," IEEE GLOBECOM, 2000.
- **Reference 3:** W. Zhang et al., "Space-Time-Frequency Coding for MIMO-OFDM in Next Generation Broadband Wireless Systems," IEEE Wireless Communications, 2008.
- **Reference 4:** D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*, Cambridge University Press, 2005.

Thank You